Assignment No-03

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Course : CSE251

Section : 22

Ans to the Q No-1

(a) $V_s(t) = 7\sin(400\pi t)$ here, $\omega = 400\pi$

We know, $\omega = 2\pi f_i$

$\therefore f_i = \frac{400\pi}{2\pi}$

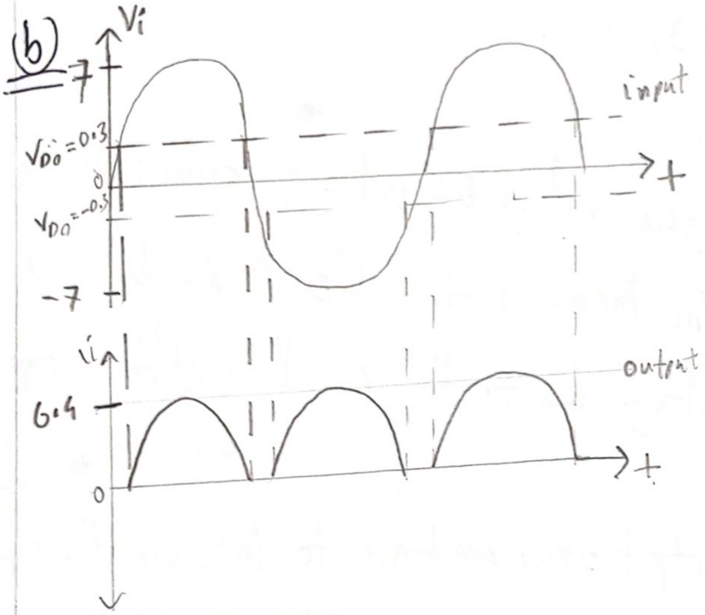
$\therefore f_i = 200$ Hz [input frequency]

For full wave rectifier, $f_r = 2 f_i$

So, output frequency $= (2\times 200)$ Hz $= 400$ Hz.

(Ans)

(b)



$V_p = V_m - 2V_{D0}$ [Full wave]

$= 7 - (2\times 0.3)$

$= 6.4$ V.

(c) $V_{DC} = \frac{2V_m}{\pi} - 2V_{D0}$

$= \frac{2\times 7}{3.1416} - (2\times 0.3)$

$= 3.85$ V.

(Ans)

$V_m = 7$ V

$V_{D0} = 0.3$ V

(d) We know, $V_{r(p-p)} = \dfrac{V_P}{f_r RC}$

$V_P = 6.4$ V
$f_r = 400$ Hz
$R = 5\times10^3\ \Omega$
$C = 100\times10^{-6}$ F

$\therefore V_{r(p-p)} = \dfrac{6.4}{400\times5\times10^{3}\times100\times10^{-6}}$ V

$= 0.032$ V. (Ans)

(e) Here, $V_{DC} = V_P - \dfrac{V_{r(p-p)}}{2}$

$V_P = 6.4$ V
$V_{r(p-p)} = 0.032$ V;
from 'c' $V_{DC} = 3.85$ V.

$= 6.4 - \dfrac{0.032}{2}$

$= 6.384$ V

Difference between $V_{DC} = (6.384 - 3.85)$ V

$= 2.534$ V.

So, it is clear that we get higher valued DC voltage with capacitor than without capacitor. The more V_{DC} increases the better it is because V_{DC} is much closer to the peak voltage due to the smoothing effect of capacitor. (Ans)

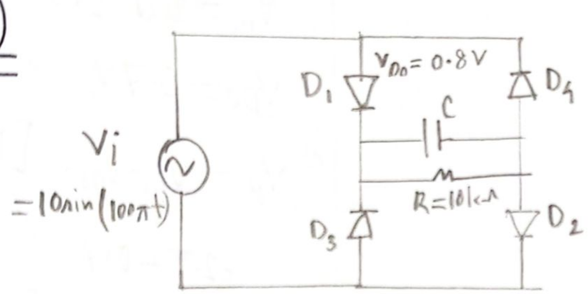
(f) For better filtering for the output waves we have to increase the capacitance of the capacitor. By doing so we will increase the τ ($\tau = RC$) of the capacitor. It will make more time for the capacitor to discharge as it can store more charge. So, the output ripple voltage will become smoother and more straight and $V_{r(p-p)}$ will be less and our V_{DC} will increase.

(g) The frequency of the ripple voltage for a full-wave rectifier is equal to output frequency. So, $f_{ripple} = f_{output} = 400\ Hz.$

Ans)

Ans to the Q No-2

(a)



(b) Full wave rectifier,

$V_i = 10\sin(100\pi t)$

$V_m = 10\ V$

$V_{D0} = 0.8\ V$

$V_p = V_m - 2V_{D0}$

$= 10 - (2 \times 0.8)\ V = 8.4\ V.$

$V_r = V_p \times 3\% = 8.4 \times \frac{3}{100}\ V = 0.252\ V.$ (Ans)

(c) $V_{DC} = V_p - \frac{V_{r(p-p)}}{2}$

$V_p = 8.4\ V$

$V_{r(p-p)} = 0.252\ V.$

$= 8.4 - \frac{0.252}{2}$

$= 8.274\ V.$ (Ans)

(d) We know,

$f_r = 2 \times 50 = 100\ Hz.$

$V_p = 8.4\ V$

$R = 10 \times 10^3\ \Omega$

$V_r = 0.252\ V$

$V_r = \frac{V_p}{f_r R C}$

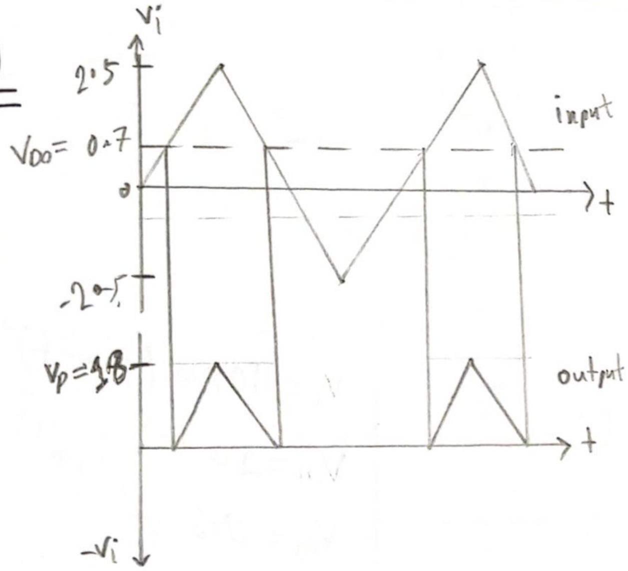
$\therefore C = \frac{V_p}{f_r R V_r}$

$$\therefore C = \frac{8.4}{100 \times 10 \times 10^{3} \times 0.252} = 33.33\ \mu F$$

$\therefore C = 33.33\ \mu F$ <u>Ans)</u>

Ans to the Q No-3

(a)



$V_m = 2.5$

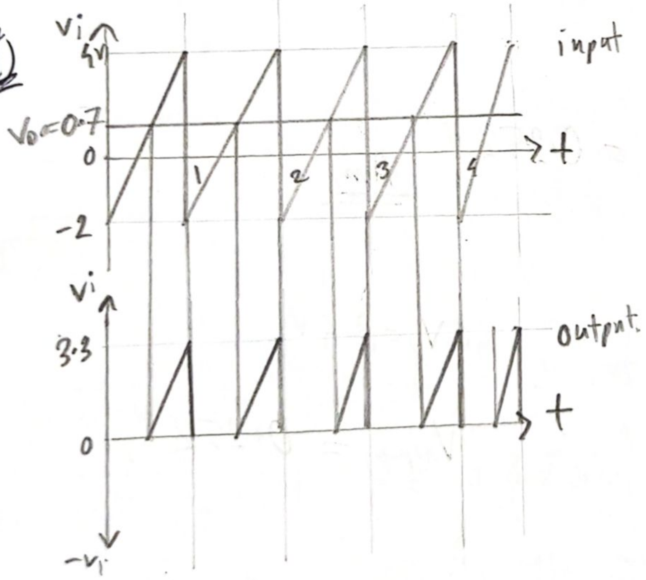
$V_{D0} = 0.7\ V$

$V_p = V_m - V_{D0}$ [Half wave]

$= 2.5 - 0.7$

$= 1.8$

(b) (i)

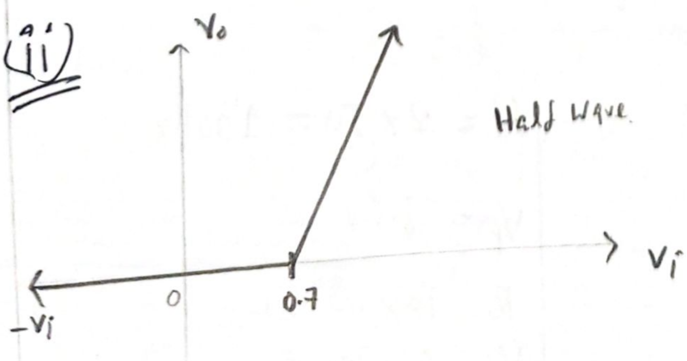


$V_m = 4V.$

$V_p = V_m - V_{D0}$

$= 4 - 0.7 = 3.3\ V.$

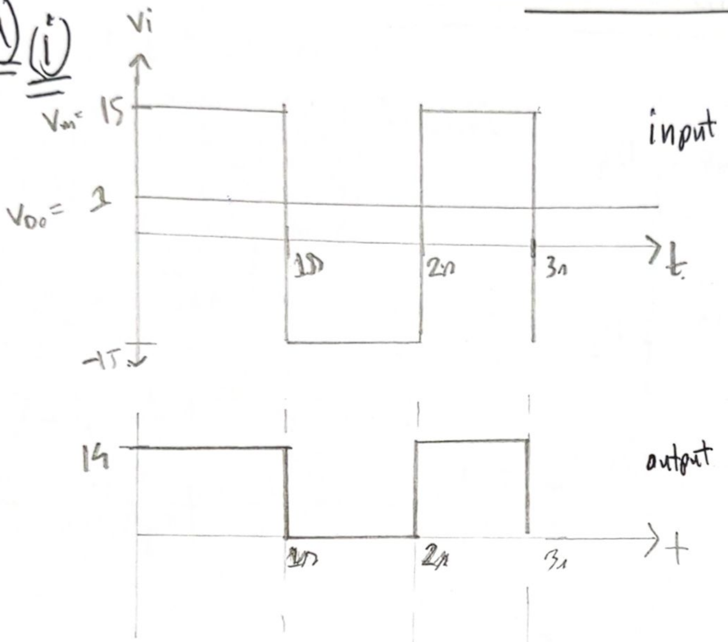
(ii)



Half wave.

Ans to the Q No-4

(a) (i)

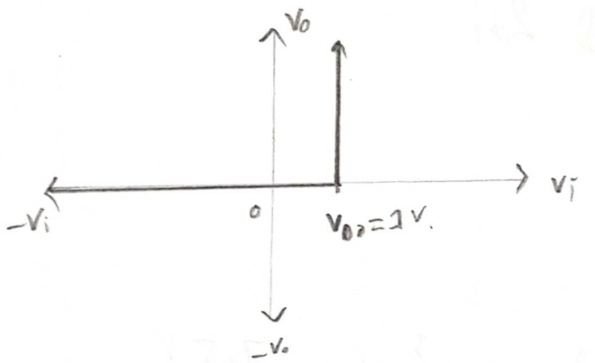


$f_i = 0.5\ Hz$

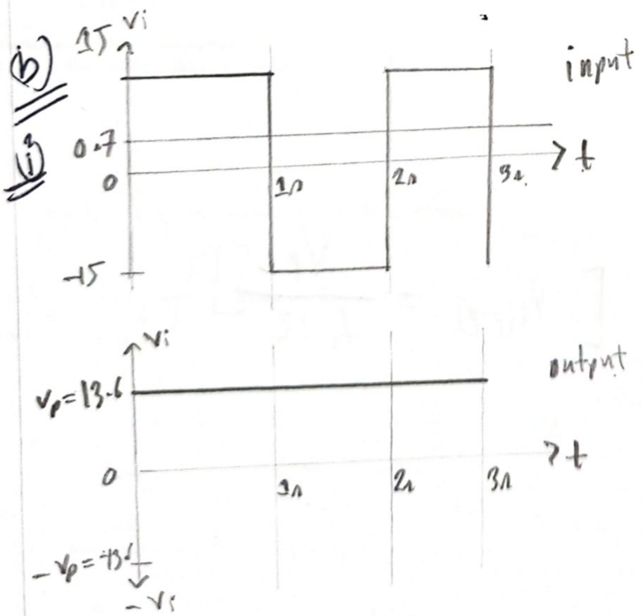
$T = \frac{1}{f_i} = \frac{1}{0.5} = 2s.$

$V_p = V_m - V_{Do} = 15 - 1 = 14V.$

(ii)



(b) (i)

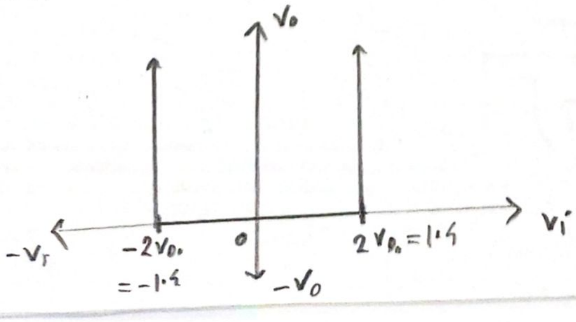


$f_i = 0.5\ Hz$

$T = \frac{1}{f_i} = \frac{1}{0.5} = 2s$

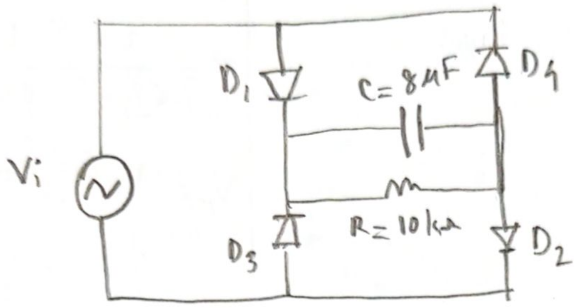
$V_p = V_m - 2V_{Do} = (15 - 2 \times 0.7) = 13.6V$

(ii)



Ans to the Q No-5

(a)



$V_{D_0} = 7V$

(b) $V_i = V_m \sin(\omega t)$

$f_r = 100\,Hz$

For full wave rectifier, $f_r = 2f_i$

$f_i = \frac{100}{2}\,Hz = 50\,Hz.$

$\omega = 2\pi f_i = 100\pi.$

Now, $V_{DC} = I_{DC} \cdot R = 0.75\times10^{-3} \times 10\times10^{3}\,V = 7.5\,V.$

We know,

$$V_{DC} = V_p - \frac{V_{r(p-p)}}{2}$$

$\left[\because V_{r(p-p)} = \frac{V_p}{f_r RC}\right]$

$$\Rightarrow 7.5 = V_p - \frac{1}{2}\left(\frac{V_p}{f_r RC}\right)$$

$$\Rightarrow V_p\left(1 - \frac{1}{2f_r RC}\right) = 7.5$$

$$\Rightarrow V_p = \frac{7.5}{\left(1 - \frac{1}{2\times100\times10\times10^{3}\times8\times10^{-6}}\right)}$$

$$\Rightarrow V_p = 8\,V.$$

(1)

We know,

$$V_p = V_m - 2V_{D0}$$

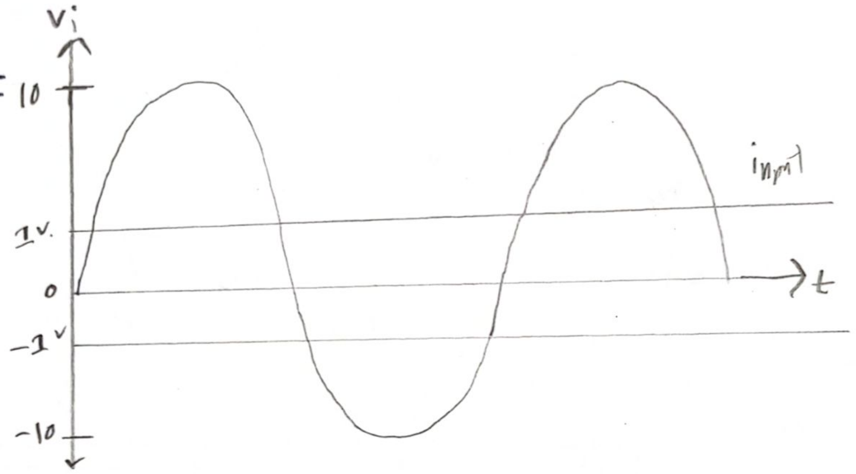
$$\therefore V_m = V_p + 2V_{D0}$$

$$\therefore V_m = 8 + 2$$

$$V_m = 10V$$

So, input waveform, $V_i = 10 \sin(100\pi t)$ (Ans)

(c)



$V_{D0} = 1V$